

AirMush

SMART URBAN INSTALLATION

START PURIFICATION



AIRMUSH
BREATHING AIR

Air Quality in Turin

Turin's Po Valley position results in continuous air pollution since the mountains hold still air and limit lateral dispersion. The primary sources of emissions that affects negatively the local air quality are road traffic, house heating, and biomass burning; regional transit from the broader Po Valley and occasional Saharan dust also play a role.

MAIN POLLUTANTS							
AQI				$\mu\text{g}/\text{m}^3$			
83	38	50	8	24	33	25	24
PM2.5	PM10	NO2	O3	PM2.5	PM10	NO2	O3

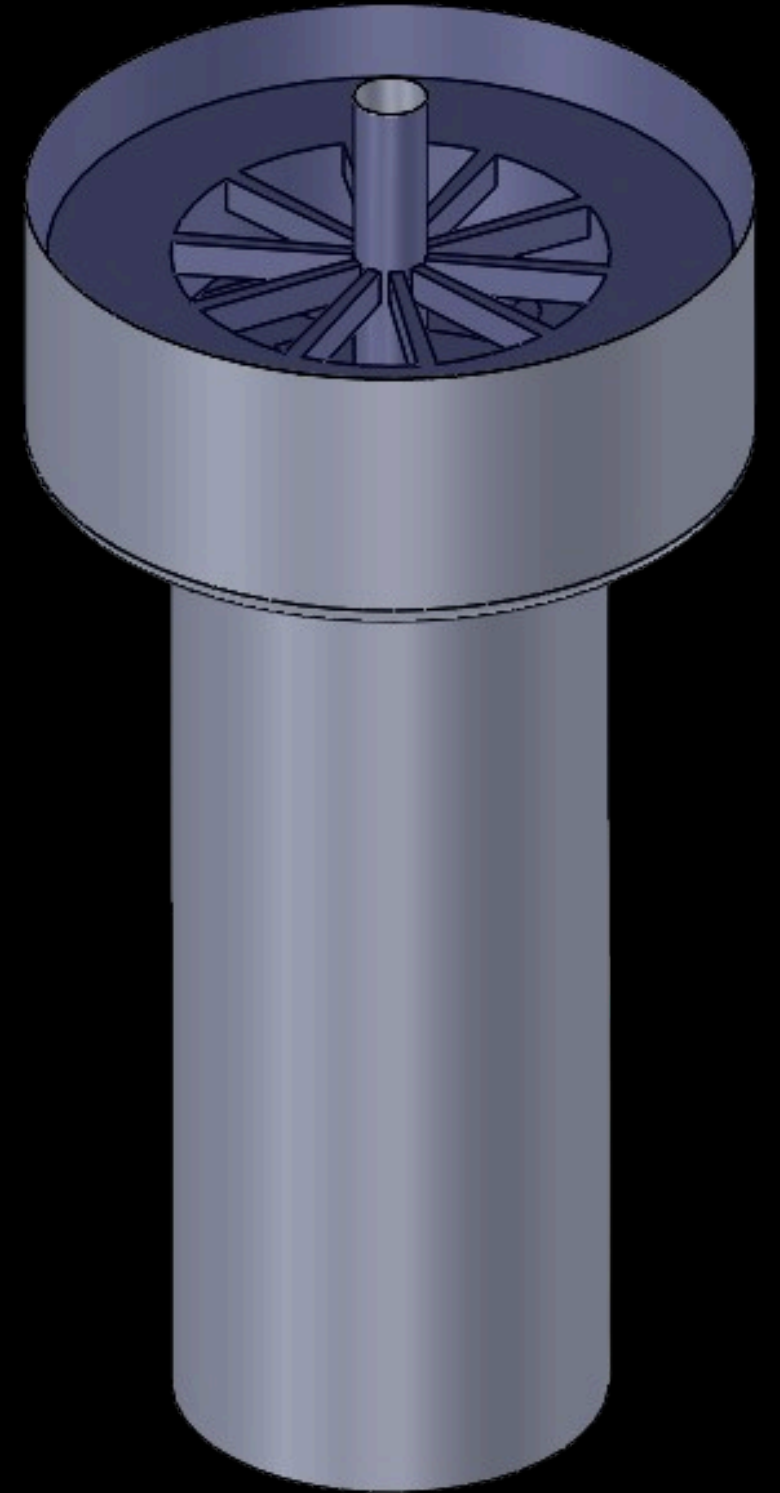


Introduction to AirMush

innovative air purifier using sustainable local bio-filters

Product concept:

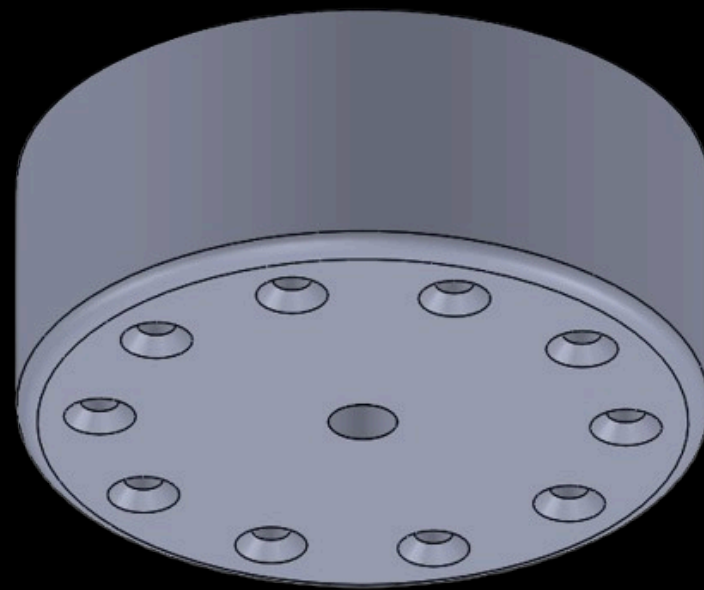
- tower shaped two piece structure, consisting of a body and upper purification center
- spherical solar panel on top(captures light more evenly from different angles)-main energy source
- bio-filters
- telecommunication tools-antenna, GPS module
- PM_{2.5} / PM₁₀ Laser Scattering Sensors,gas sensors, standard sensor package(temperature,humidity, barometric), UV sensors and etc
- use of aluminum and carbon steel materials (resilience and anti-corrosion properties)



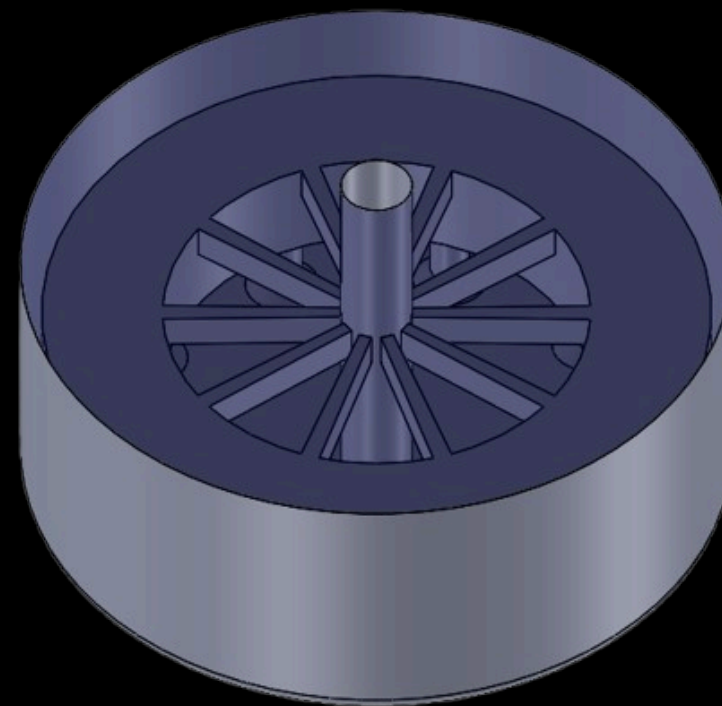
Purifying mechanism

CYCLOTRON PRINCIPLE:

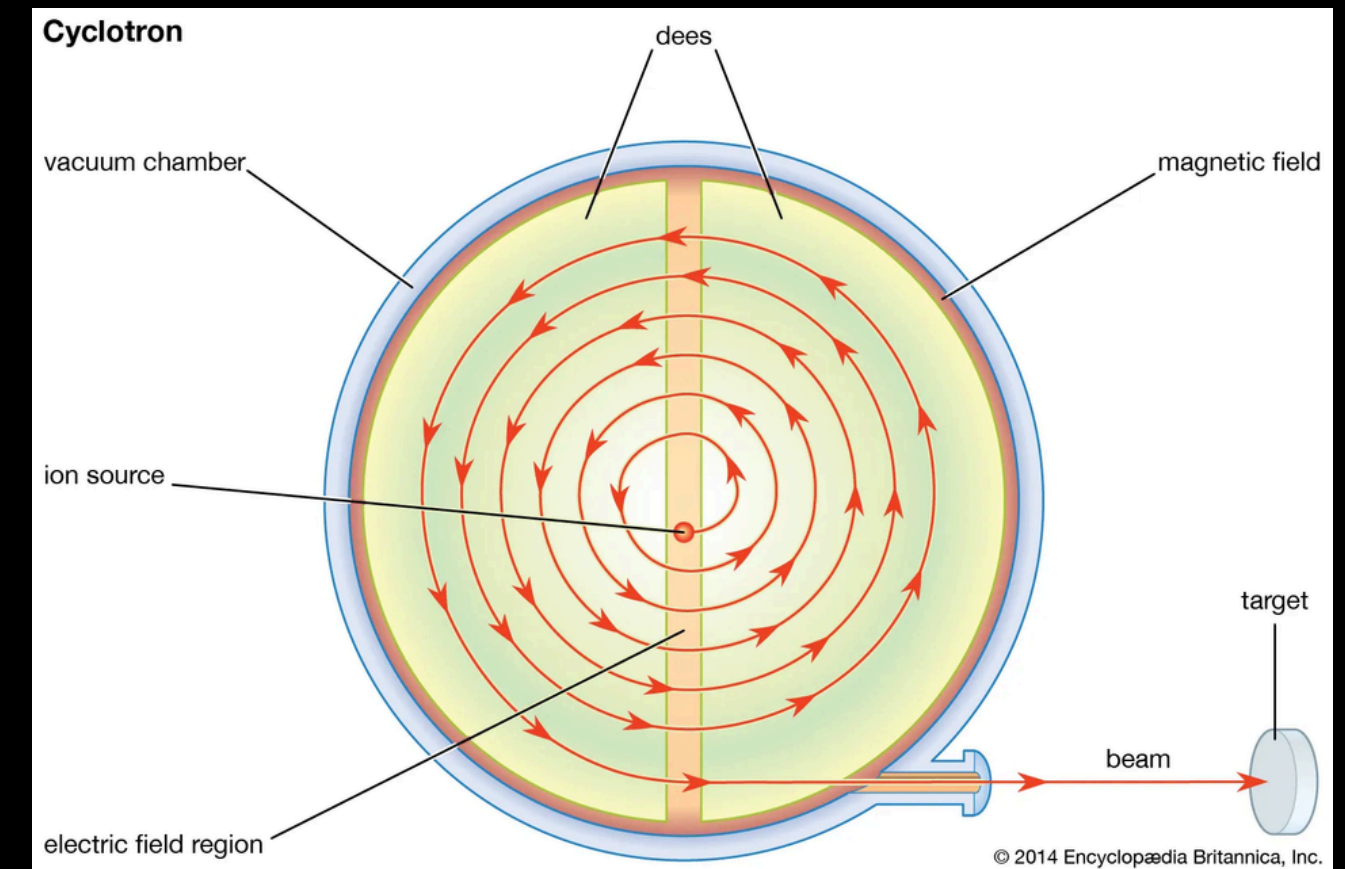
Air enters through the outer holes of the structure. The particles are situated in the center of the cyclotron. An electric field, powered by the spherical solar panel accelerates them, whilst the magnetic field contributes to the change of direction. Inside the cyclotron there are chambers with different **sustainable natural bio-filters**. Air particles circulate until fully purified and exit the device from single tube on top with greater speed.



bottom-view



top-view



BIOFILTERS:

WHAT ARE BIOFILTERS?

They are systems that use organic materials to remove pollutants from air.

HOW BIOFILTERS WORK?

Dirty air flows through a filter made of natural materials like moss, coconut husk, or clay.

Microbes and plant roots inside the filter break down harmful gases like VOCs and NOx.

The result is clean, fresh air.

WHY DID WE CHOOSE THEM?

- They work passively and continuously
- Require little or no maintenance
- Fit easily inside modular airflow paths
- Are sustainable and eco-friendly.



Why **AirMush**?

SUSTAINABLE AND ECOLOGICAL:



- entire use of green energy(solar and nuclear)
- efficient,easily produced and eco-friendly materials
- bio-filters based of natural and organic ingredients from the Pienmonte region
- scaled and replicable



INNOVATIVE AND TECHNOLOGICAL:

- integration of digital components
- data visialization and information exchange with citizens(social)
- AI agents
- environmental monitoring
- implemantation of fundamental particles power

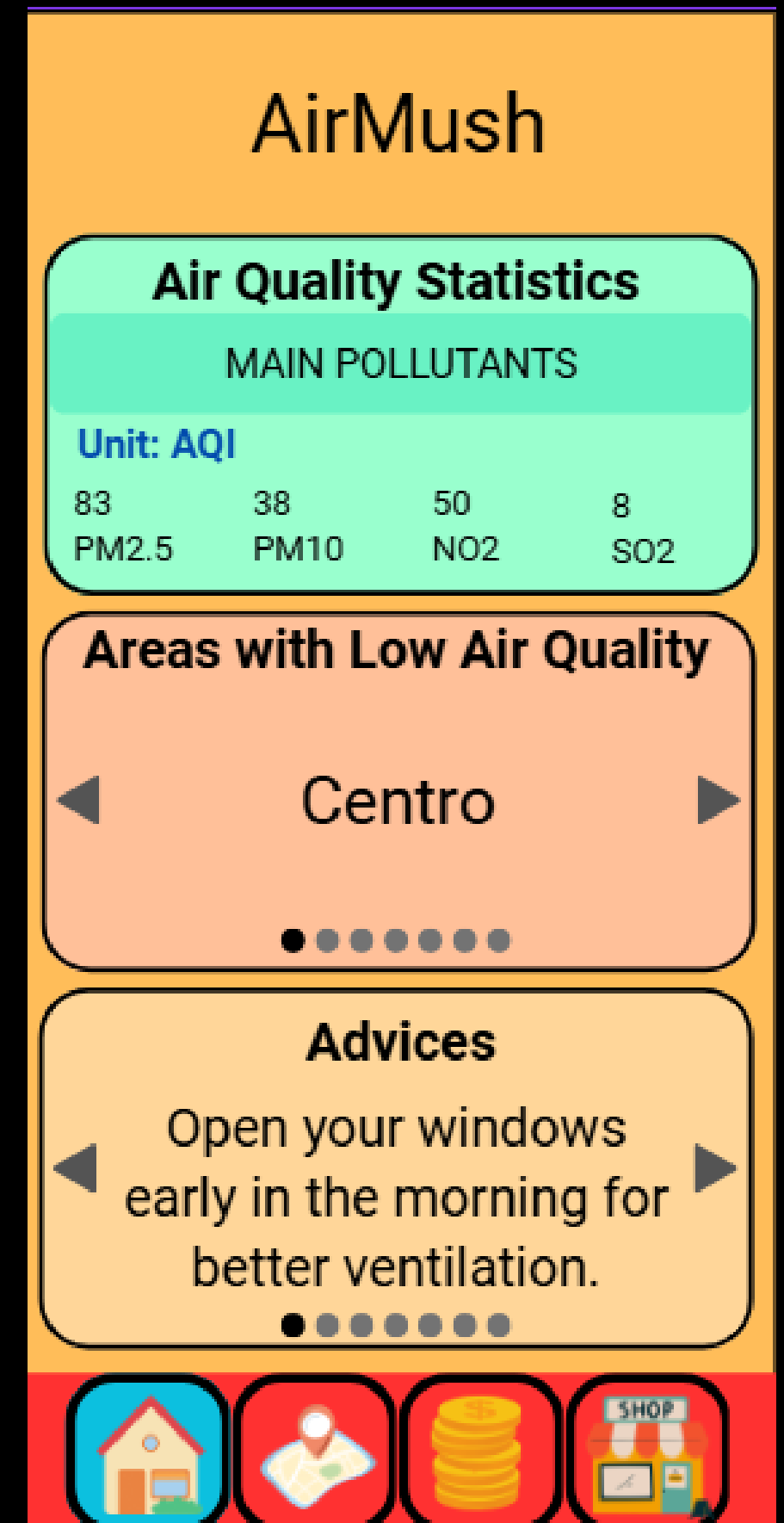
AirMush represents the perfect intersection of innovation and environmental preservation.

Due to its easy meintenance and production , use of bio-materials and sustainable energysources, it decreases the unnecessary waste of resources and makes our town a better place to live.

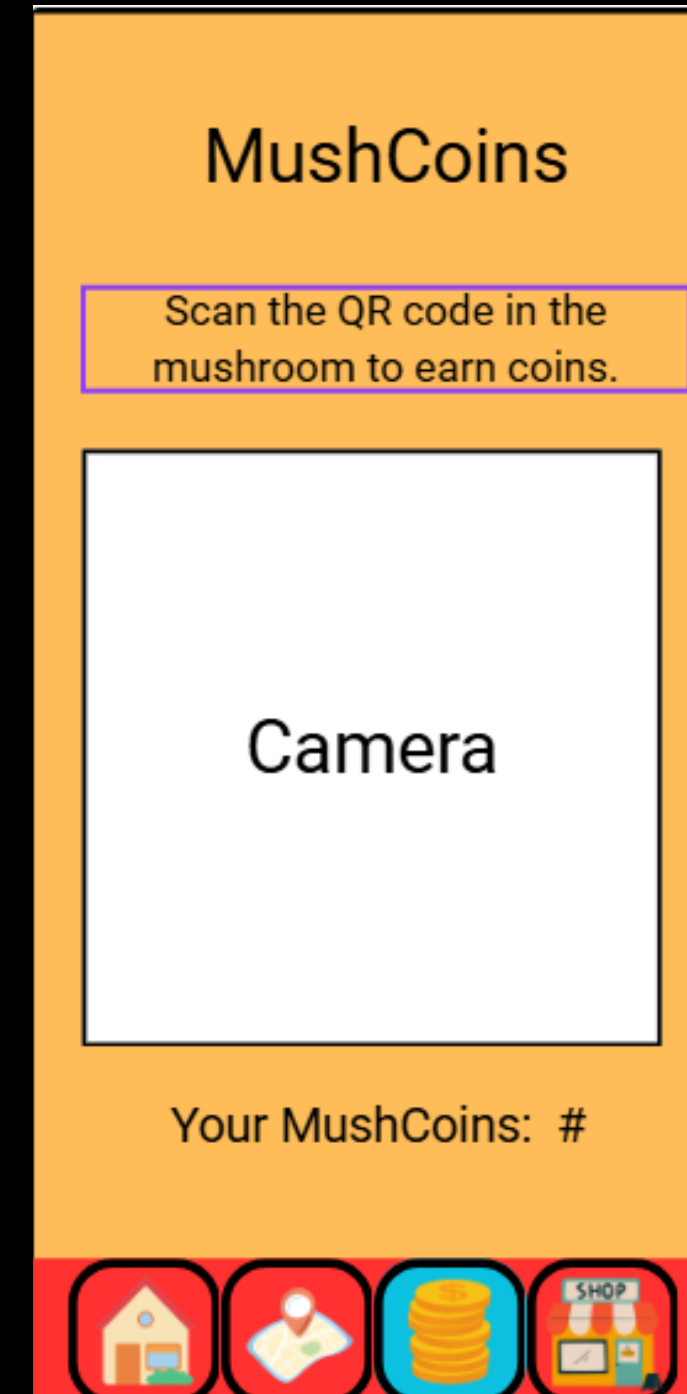
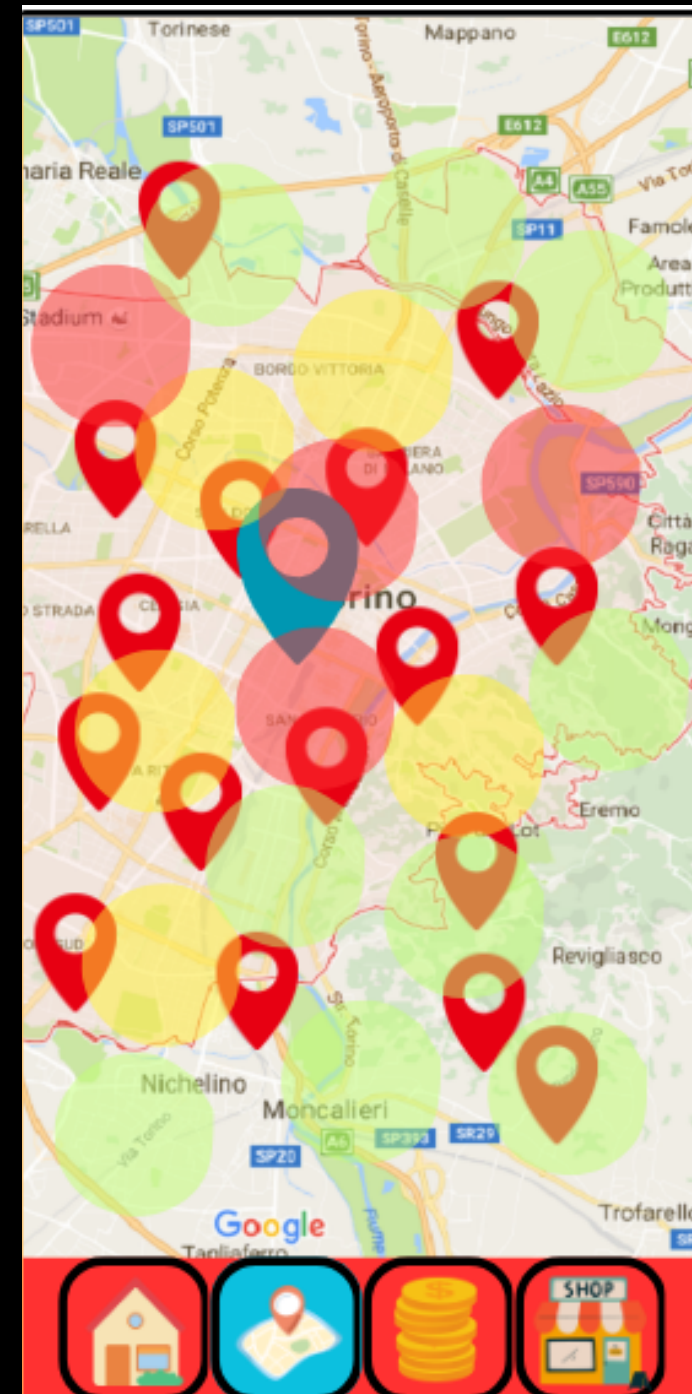
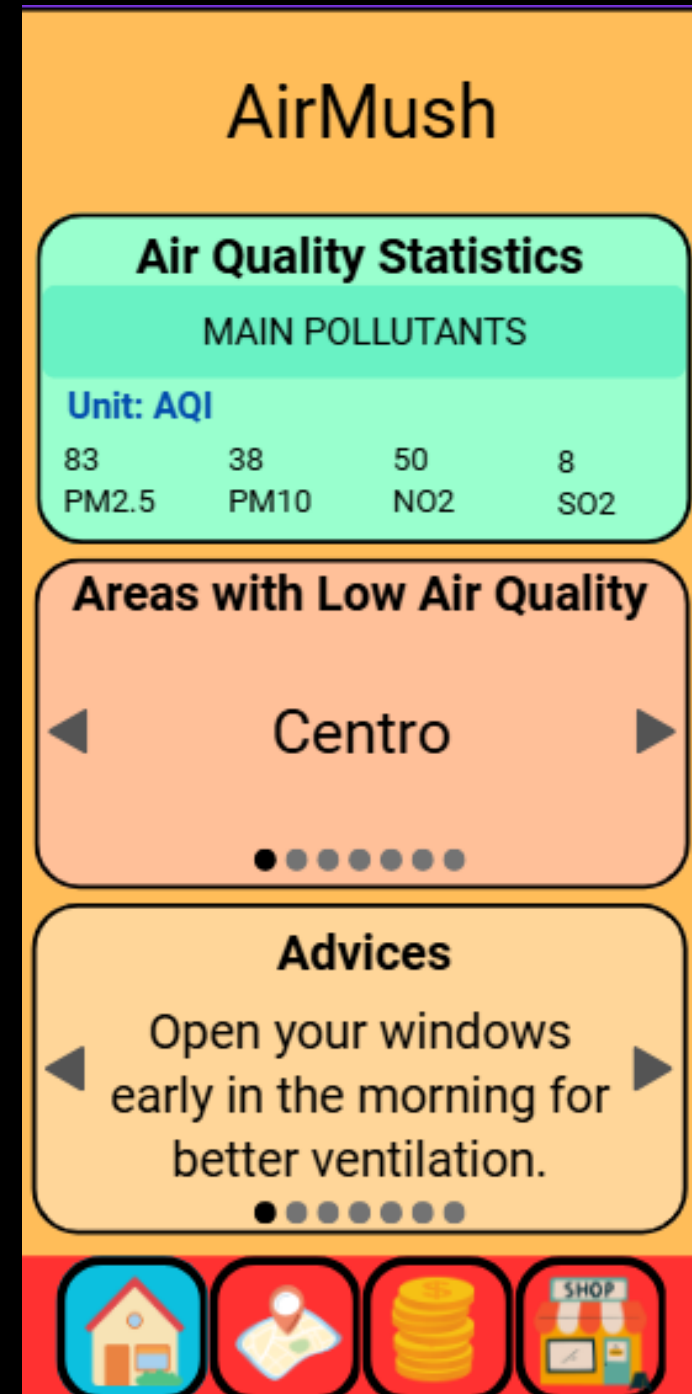
AirMush Application

AirMush app will be available to citizens of Turin to download. Through the app, they'll learn about air pollution and they'll know the statistics about the air quality in Turin. The air mushrooms will send their sensor data to a big mother air mushroom that contains all the data from other small mushrooms.

This mother mushroom will provide all the information on the app. Other mushrooms will connect to this mother mushroom via telecommunication. If the air around a small mushroom is clean enough, the mother mushroom will deactivate the small mushroom to avoid wasting energy.



AirMush App Pages



The home page will contain statistics coming from the mother mushroom and also it will warn citizens about areas with low air quality. To raise awareness, it will also give advices to the users. In the map page, the users will see the map of Turin with the locations of mushrooms. The map will also show the quality of air through colors. When users scan the QR codes in the mushrooms, they will gain coins which they can spend to earn discounts from local businesses.